Verification and Validation Report: Natural Control

Group 7
Lucas DeBoer
Kevin Kim
Aaron Loh
Curtis Vos

October 19, 2026

1 Revision History

Date	Version	Notes
Date 1	1.0	Notes
Date 2	1.1	Notes

2 Symbols, Abbreviations and Acronyms

symbol	description
T	Test

[symbols, abbreviations or acronyms – you can reference the SRS tables if needed: SS]

Contents

1 Revision History **i**

2 Symbols, Abbreviations and Acronyms **ii**

3 Functional Requirements Evaluation **1**

4 Nonfunctional Requirements Evaluation **1**

4.1 Usability 1

4.2 Performance 1

4.3 etc. 1

5 Comparison to Existing Implementation **1**

6 Unit Testing **1**

7 Changes Due to Testing **1**

8 Automated Testing **1**

9 Trace to Requirements **2**

10 Trace to Modules **2**

11 Code Coverage Metrics **2**

12 Custom Document Tasks **2**

13 Appendix: Generative AI Use Disclosure **3**

13.1 AI Tools Used 3

13.2 How and Where Used 3

13.3 Verification of AI-Generated Content 3

14 Appendix: Reflection **4**

List of Tables

List of Figures

This document ...

3 Functional Requirements Evaluation

4 Nonfunctional Requirements Evaluation

4.1 Usability

4.2 Performance

4.3 etc.

5 Comparison to Existing Implementation

This section will not be appropriate for every project.

6 Unit Testing

7 Changes Due to Testing

[This section should highlight how feedback from the users and from the supervisor (when one exists) shaped the final product. In particular the feedback from the Rev 0 demo to the supervisor (or to potential users) should be highlighted.: SS]

8 Automated Testing

[The automated testing section summarizes the team's experience with automated testing. You can include a brief recap of your approach to automated testing, along with a pointer to details in the VnV plan. The VnV report should capture what was learned and what was changed during the course of the project. That means that this section can be short if there wasn't any need to modify the plan, or if everything went well. If there were problems, or changes, or planned changed, they should be mentioned here.: SS]

9 Trace to Requirements

10 Trace to Modules

11 Code Coverage Metrics

[The code coverage metrics summarize your code coverage. Tools to measure statement coverage are relatively common. Your team should summarize the coverage results from your tool. It is unlikely that 100% coverage will be achieved. A discussion of any theories on why 100% was not achieved would be appropriate. Ideas on who to improve the coverage would also be welcome.: SS]

12 Custom Document Tasks

[Summary of the status of your custom document tasks. If feasible, both custom document tasks should be complete by the time of writing the VnV report. If they are not complete, you should summarize the plans for completing them.: SS]

13 Appendix: Generative AI Use Disclosure

[ChatGPT (version 5) was used to brainstorm ideas for the template for this disclose section.: SS]

13.1 AI Tools Used

[Give the name of the AI tool(s) used and the version.: SS]

13.2 How and Where Used

[Explain what the tool(s) were used for. Examples include: brainstorming, explaining a technical concept, reviewing the document, generating or modifying text, generating or modifying diagrams, identifying errors or omissions or inconsistencies.: SS]

[For each of the uses, identify which parts of the document were affected.: SS]

[You do not need to report every interaction, but you should disclose the uses that materially contributed to your work: SS]

13.3 Verification of AI-Generated Content

[

- Describe how the team checked AI-generated or AI-assisted material before including it in the document.
- In particular, verify factual claims, technical assertions, requirements, calculations, references, and other information for which an AI system may produce incorrect or fabricated results.
- **The team is responsible for the accuracy, completeness, consistency, and appropriateness of the submitted document, regardless of whether material was generated by a human or an AI system.**

: SS]

14 Appendix: Reflection

The information in this section will be used to evaluate the team members on the graduate attribute of Reflection.

The purpose of reflection questions is to give you a chance to assess your own learning and that of your group as a whole, and to find ways to improve in the future. Reflection is an important part of the learning process. Reflection is also an essential component of a successful software development process.

Reflections are most interesting and useful when they're honest, even if the stories they tell are imperfect. You will be marked based on your depth of thought and analysis, and not based on the content of the reflections themselves. Thus, for full marks we encourage you to answer openly and honestly and to avoid simply writing "what you think the evaluator wants to hear."

Please answer the following questions. Some questions can be answered on the team level, but where appropriate, each team member should write their own response:

1. What went well while writing this deliverable?

2. What pain points did you experience during this deliverable, and how did you resolve them?

3. Which parts of this document stemmed from speaking to your client(s) or a proxy (e.g. your peers)? Which ones were not, and why?

4. In what ways was the Verification and Validation (VnV) Plan different from the activities that were actually conducted for VnV? If there were differences, what changes required the modification in the plan? Why did these changes occur? Would you be able to anticipate these changes in future projects? If there weren't any differences, how was your team able to clearly predict a feasible amount of effort and the right tasks needed to build the evidence that demonstrates the required quality? (It is expected that most teams will have had to deviate from their original VnV Plan.)

5. Reflect on your use of CI for continuous integration testing (regression testing). Did your CI work to gatekeep your PRs? Would it have been helpful if you got your CI working earlier in the project? What are your plans to use CI for testing in future projects?